

Learning Outcomes

At the end of the session, you will be able to:

- Understanding linear regression
- Create a relationship model using the lm() functions in R
- To use the predict() function in R and create regression plot
- Performing K Nearest Neighbor on Dataset

Activity

1. Using the built-ins dataset "Theoph", create a linear model and visualize the linear regression. Discuss the linear model summary. Predict the dose of theophylline administered orally to the subject (mg/kg) if the weight are 90kg, 95kg and 100kg.

```
x=Theoph$Wt
y=Theoph$Dose
model = lm(y~x)
plot(x,y,col = "blue",main = "Weight & Dose Regression",
      abline(lm(y~x)),
      cex = 1.3,pch = 16,
      xlab = "Weight of the subject (kg)",
      ylab = "Dose of Theophylline (mg/kg)")

print(summary(model))

newweight <- data.frame(x=c(90, 95, 100))
result <- predict(model,newweight)
print(result)
```

2. Using the built-ins dataset "ChickWeight" , perform the KNN algorithm. Find the optimal K value for which experimental diet the chick received (column=Diet). Discuss the confusion matrix achieved with the optimal K value.

```
# Loading package
library(e1071)
library(caTools)
library(class)

# Loading data
data(ChickWeight)
head(ChickWeight)

# Splitting data into train and test data
split <- sample.split(ChickWeight, SplitRatio = 0.8)
train_cl <- subset(ChickWeight, split == "TRUE")
test_cl <- subset(ChickWeight, split == "FALSE")

# Feature Scaling
train_scale <- scale(train_cl[, 1:2])
test_scale <- scale(test_cl[, 1:2])
```

```
acc_list = 0

for (x in 1:200) {
  classifier_knn <- knn(train = train_scale,
                        test = test_scale,
                        cl = train_cl$Diet,
                        k = x)

  misClassError <- mean(classifier_knn != test_cl$Diet)
  acc = 1-misClassError
  print(paste(x,': Accuracy =', acc))
  acc_list = append(acc_list, acc)
}

optimal_k = which(acc_list == max(acc_list))

classifier_knn <- knn(train = train_scale,
                      test = test_scale,
                      cl = train_cl$Diet,
                      k = optimal_k)

misClassError <- mean(classifier_knn != test_cl$Diet)
acc = 1-misClassError
print(paste('Highest Accuracy =', acc, ' with K = ',optimal_k))

# Confusion Matrix
cm <- table(test_cl$Diet, classifier_knn)
print(cm)
```

Any other logical method to achieve optimal K value.

Submission

- Submit to your GA.